Learning Objective 3.2: I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Reading a sweep.** A lab group sweeps the PHASER with all eight elements active and a source at boresight. Working at 10.3 GHz with element spacing $d = 14$ mm, they record the trace below. The sweep step is $2.8125°$ and the peak reads -4.1 dBFS.

θ (°)	0.0	2.8	5.6	8.4	11.3	14.1	16.9	19.7	22.5	25.3
Power (dBFS)	-4.1	-4.6	-6.2	-9.1	-14.3	-24.1	-22.0	-17.4	-16.6	-18.1

(a) Using linear interpolation between the two samples that bracket the half-power level, find the half-power beamwidth. The trace is symmetric about $0°$.

$\theta_{\text{HP}} =$ $12.9°$

Half power is 3 dB below the peak, so the level is $-4.1 - 3.0 = -7.1$ dBFS. That falls between the samples at $5.6°$ (-6.2) and $8.4°$ (-9.1).

$$\theta_{-3} = 5.6 + 2.8\left(\frac{7.1 - 6.2}{9.1 - 6.2}\right) = 5.6 + 2.8(0.31) = 6.5°$$

The trace is symmetric, so the beamwidth is twice this.

$$\theta_{\text{HP}} = 2(6.5) = 12.9°$$

Theory gives $13.2°$, so the sweep reads about three tenths of a degree narrow.

(b) Find the first-null beamwidth as the sweep reports it, and compare it with FNBW $= 2\arcsin(\lambda/Nd)$.

$\text{FNBW}_{\text{meas}} =$ $28.2°$

$\text{FNBW}_{\text{calc}} =$ $30.1°$

The deepest sample between the main lobe and the sidelobe is -24.1 dBFS at $14.1°$, so the sweep reports

$$\text{FNBW}_{\text{meas}} = 2(14.1) = 28.2°$$

With $\lambda = 29.1$ mm and $Nd = 8(14) = 112$ mm,

$$\text{FNBW} = 2\arcsin\left(\frac{29.1}{112}\right) = 2\arcsin(0.2598)$$
$$= 2(15.1°) = 30.1°$$

The measured value is $1.9°$ narrow because the true null at $15.1°$ falls between the samples at $14.1°$ and $16.9°$. The sweep can only report a null at an angle it actually visited.

(c) Find the first sidelobe level relative to the peak, in dBc.

SLL = -12.5 dBc

The tallest sample beyond the null is -16.6 dBFS at $22.5°$. Relative to the peak,

$$\text{SLL} = -16.6 - (-4.1) = -12.5 \text{ dBc}$$

which is within a few tenths of the -12.8 dB the array factor gives for eight uniform elements.

(d) No sample in the sweep reads below about -24 dBFS, which is 20 dB below the peak, yet the noise

floor on this kit is 23 dB below the peak. Explain in one or two sentences why the deepest sample sits 3 dB above the floor rather than on it.

The receiver reports signal power plus noise power. At the sample nearest the null the array factor has fallen to about -23 dB, which is the floor level itself, so the two contributions are equal and their sum is 3 dB above either one. A measured null depth is therefore a statement about the dynamic range of the measurement rather than about the array.

2. **Predict before you measure.** The same array is reconfigured by setting some element gains to zero. The spacing $d = 14$ mm and the frequency 10.3 GHz do not change, so $\lambda = 29.1$ mm throughout.

   (a) Rx1, Rx2, Rx7 and Rx8 are set to zero, leaving Rx3 through Rx6 active — four adjacent elements. Predict the half-power beamwidth from $\theta_{\text{HP}} \approx 0.886\lambda/(Nd)$.

   $\theta_{\text{HP}} =$ ≈ 26-27°

   Four active elements span $Nd = 4(14) = 56$ mm.

   $$\theta_{\text{HP}} \approx \frac{0.886(29.1)}{56} = 0.4604 \text{ rad} = 26.4^\circ$$

   Solving the array factor exactly gives 27.4°; the small-angle step inside the 0.886 rule costs about a degree at this beamwidth.

   (b) Now only Rx3, Rx4 and Rx5 are left on — three adjacent elements. Predict the half-power beamwidth and the first-null beamwidth.

   $\theta_{\text{HP}} =$ ≈ 35°

   FNBW = 87.7°

   Three adjacent elements span $Nd = 3(14) = 42$ mm.

   $$\theta_{\text{HP}} \approx \frac{0.886(29.1)}{42} = 0.6138 \text{ rad} = 35.2^\circ$$

   $$\text{FNBW} = 2\arcsin\left(\frac{29.1}{42}\right) = 2\arcsin(0.6929)$$
   $$= 2(43.8^\circ) = 87.7^\circ$$

   The beam is wide enough that the 0.886 rule is starting to under-predict; expect the sweep to read closer to 37°.

   (c) By how many dB should the peak of the three-element sweep sit below the peak of the full eight-element sweep, if every element gain that is still on remains at 100%?

   $\Delta =$ −8.5 dB

   At boresight the element signals add in voltage, so the peak field is proportional to the number of active elements.

   $$\Delta = 20\log_{10}\left(\frac{3}{8}\right) = 20\log_{10}(0.375) = -8.5 \text{ dB}$$

   (d) A student proposes reaching the same 35° beamwidth by leaving all eight elements on and turning every gain down to 37.5%. Explain in one or two sentences why that does not work.

   Beamwidth is set by the active aperture Nd, not by how much gain the elements are given. Turning all eight gains down uniformly keeps the aperture at 112 mm, so the beam stays at about 13° and only the peak level drops. To widen the beam the aperture itself has to shrink.

3. **Reconciling a sidelobe.** A second group measures the first sidelobe of the full eight-element array at -11.5 dBc, while the array factor for $N = 8$ predicts -12.8 dB. Their noise floor sits at -23 dBc and the sweep step is $2.8125°$.

   (a) The floor adds noise power to the sidelobe power the receiver reports. Treating the two as uncorrelated, compute the level the sweep should report for a true -12.8 dB sidelobe sitting on a -23 dBc floor.

   SLL = -12.4 dBc

   Convert both to power ratios and add.

   $$p_{\text{sig}} = 10^{-12.8/10} = 0.0525, \qquad p_{\text{n}} = 10^{-23/10} = 0.0050$$

   $$p_{\text{tot}} = 0.0525 + 0.0050 = 0.0575$$

   $$\text{SLL} = 10\log_{10}(0.0575) = -12.4 \text{ dBc}$$

   The floor alone accounts for about 0.4 dB of lift.

   (b) Their peak reading is also affected. The peak is 23 dB above the floor by definition. By how much does the floor lift the peak reading, and which way does that push the reported sidelobe level in dBc?

   peak lift = $+0.02$ dB

   Normalize the peak signal power to 1.

   $$p_{\text{tot}} = 1 + 0.0050 = 1.0050$$
   $$10\log_{10}(1.0050) = +0.02 \text{ dB}$$

   The peak lifts by only about 0.02 dB, which is negligible. Because the sidelobe lifts by 0.4 dB and the peak barely moves, the ratio of the two moves *up*, making the reported sidelobe look higher than the array factor predicts.

   (c) The floor explains 0.4 dB of the 1.3 dB discrepancy. Name two other contributions that plausibly make up the rest, and say for each whether it would move the reading up or down.

   First, run-to-run noise on the sidelobe sample itself. The sidelobe is only about 10 dB above the floor, so a single sweep reading scatters a few tenths of a decibel in either direction. Second, residual amplitude and phase errors across the eight elements after calibration. These break the uniform distribution slightly and raise the sidelobes, so that contribution moves the reading up. Grid straddle is a third contribution, but it moves the reading *down* and by very little here, since the sidelobe peak at $21.9°$ is caught by the sample at $22.5°$ to within 0.05 dB.

4. **The two-element case.** With only Rx4 and Rx5 active the array has two elements at $d = 14$ mm, and at 10.3 GHz the array factor is $AF = \cos(\psi/2)$ with $\psi = kd\sin(\theta)$.

   (a) Find the half-power beamwidth exactly, without using the $0.886\lambda/(Nd)$ rule.

   θ_{HP} = 62.6°

   Set the array factor to $1/\sqrt{2}$.

   $$\cos(\psi/2) = \frac{1}{\sqrt{2}} \quad \Rightarrow \quad \frac{\psi}{2} = \frac{\pi}{4} \quad \Rightarrow \quad \psi = \frac{\pi}{2}$$

   With $\psi = kd\sin(\theta)$ and $kd = 2\pi d/\lambda$,

   $$\sin(\theta) = \frac{\pi/2}{2\pi d/\lambda} = \frac{\lambda}{4d} = \frac{29.1}{4(14)} = 0.5196$$

   $$\theta = 31.3^\circ \quad \Rightarrow \quad \theta_{HP} = 62.6^\circ$$

   (b) Show that the $0.886\lambda/(Nd)$ rule fails here, and state the percentage error it makes.

   error = -15.7%

   $$\theta_{HP} \approx \frac{0.886(29.1)}{28} = 0.9208 \text{ rad} = 52.8^\circ$$

   $$\text{error} = \frac{52.8 - 62.6}{62.6} = -15.7\%$$

   The rule assumes the half-power point is at a small enough angle that $\sin(\theta) \approx \theta$. At 31° that is no longer true.

   (c) Evaluate $\lambda/2d$ and use it to explain why the lab table lists the first-null beamwidth of this array as 180°.

   $\lambda/2d$ = 1.039

   The first null of $\cos(\psi/2)$ needs $\psi = \pi$, which requires

   $$\sin(\theta) = \frac{\lambda}{2d} = \frac{29.1}{2(14)} = 1.039$$

   No real angle has a sine greater than 1, so the null lies outside visible space and no null appears anywhere in the $\pm 90^\circ$ scan. With no null to measure, the beamwidth between first nulls is quoted as the full 180° scan range by convention.

(d) How far below its own peak does this two-element pattern fall at $\theta = 90°$? Compare that with the 23 dB of dynamic range the sweep has above the noise floor at eight elements, and say what the trace will look like near the edges of the scan.

drop at $90°$ = -24.5 dB

At $\theta = 90°$, $\sin(\theta) = 1$ and $\psi = 2\pi(14)/29.1 = 3.023$ rad, so

$$|AF| = \cos(1.511) = 0.0597$$
$$20\log_{10}(0.0597) = -24.5 \text{ dB}$$

The two-element peak is itself 12 dB below the eight-element peak, so it clears the floor by only about 11 dB. The pattern needs 24.5 dB of range to reach endfire and the measurement has 11 dB, so the trace flattens into the noise well before $\pm 90°$ and the outer part of the roll-off is not visible.

5. **Design judgment.** Your section is asked to demonstrate a 20° beam on the PHASER at 10.3 GHz, using only the Element Gains controls.

   (a) How many adjacent elements must be active to give a half-power beamwidth closest to 20°? Give the aperture and the resulting beamwidth.

   N = 5 elements

   θ_{HP} = 21.1°

   Solve the beamwidth rule for the aperture.

   $$Nd = \frac{0.886\lambda}{\theta_{HP}} = \frac{0.886(29.1)}{20^\circ(\pi/180)} = \frac{25.78}{0.3491} = 73.8 \text{ mm}$$

   $$N = \frac{73.8}{14} = 5.3$$

   Element counts are integers, so take $N = 5$: $Nd = 70$ mm and

   $$\theta_{HP} \approx \frac{0.886(29.1)}{70} = 0.3683 \text{ rad} = 21.1^\circ$$

   Six elements would give 17.6°, which is further from the target.

   (b) What does this cost in peak received level compared with the full eight-element array, and how much dynamic range is left above the noise floor?

   Δ = -4.1 dB

   range = 18.9 dB

   $$\Delta = 20\log_{10}\left(\frac{5}{8}\right) = 20\log_{10}(0.625) = -4.1 \text{ dB}$$

   The floor does not move, so the range above it drops from about 23 dB to

   $$23 - 4.1 = 18.9 \text{ dB}$$

   The first sidelobe of a five-element uniform array is near -12 dB, which still clears the floor comfortably.

   (c) A classmate suggests instead applying a Hann taper to all eight elements, since that also widens the beam to about 19.5°. State one advantage and one disadvantage of the taper against switching elements off, and say which you would choose for a demonstration whose purpose is to show that beamwidth depends on aperture.

   The taper keeps the full 112 mm aperture and pushes the sidelobes below the noise floor, so the trace is cleaner and the main lobe is the only thing on the plot. Its disadvantage for this purpose is that it widens the beam by changing the shape of the amplitude distribution, not by changing the aperture, so it demonstrates the opposite of the intended point. It also costs about 4.7 dB of peak, which is no better than the five-element case. For a demonstration about aperture, switch elements off: the beamwidth and the aperture then move together and the claim is visible in one control.

Documentation: _____